

Explainable AI Reveals Hidden Biomarkers in Parkinson's Brain Imaging

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Abstract— Parkinson's disease (PD) is the second most common neurodegenerative disorder, characterized by progressive dopaminergic neuron loss and diverse motor and non-motor symptoms. Early and accurate diagnosis remains challenging due to the lack of definitive biomarkers. While DaTSCAN SPECT imaging supports clinical assessment, interpretation often centers on striatal regions, potentially overlooking other areas involved in disease progression. This study presents a robust deep learning approach to classify PD and investigate the role of non-striatal regions using explainable artificial intelligence (XAI). A 3D Convolutional Neural Network (3D CNN) was trained on DaTSCAN volumes from the PPMI database to distinguish controls from PD patients. Model interpretability was enhanced using Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize regions influencing predictions. The CNN model achieved high accuracy (>97%) across all stages. Grad-CAM heatmaps consistently revealed attention to cortical and subcortical areas beyond the striatum, suggesting distributed functional changes that may serve as early or complementary biomarkers. These findings highlight the importance of integrating explainable AI in neuroimaging to improve diagnostic precision.

Keywords—Parkinson's disease, artificial intelligence, convolutional neural network, DaTSCAN, Grad-CAM

I. INTRODUCTION

Parkinson's disease (PD) ranks as the second most prevalent neurodegenerative disorder globally, following Alzheimer's, and is particularly common among individuals aged 60 and above [1]. According to recent data research conducted by the Statista Research Department, there are over 830,000 people with PD in western Europe, including 120,000 in France [2]. The disease primarily targets the dopamine-producing neurons in the brain's substantia nigra, leading to a range of motor and non-motor symptoms, such as tremors, bradykinesia, rigidity, and balance issues [3]. While the exact cause of PD remains unclear, it is widely believed to result from a complex interplay of genetic and environmental factors, with certain mutations linked to the disease in some cases [4]. However, diagnosing PD remains challenging, as the disease shares symptoms with other neurodegenerative conditions, and current methods often rely heavily on clinical evaluation and the patient's response to treatment, making early diagnosis difficult [2].

Due to the absence of a definitive laboratory test [5], diagnosing PD typically occurs in the later stages, which can delay treatment and worsen the patient's condition; that's why the diagnosis of PD is increasingly supported by non-invasive imaging techniques like SPECT and MRI scans, which offer critical insights into brain structure and function [2]. SPECT-based diagnostic tools, particularly using I123-Ioflupane (DaTSCAN), have become widely adopted in clinics and

show promise in identifying PD, especially in the early stages [6]. However, interpreting SPECT images remains complex, often requiring expert clinicians to evaluate and correlate multiple parameters, a process that is vulnerable to human error [7]. This issue has led to interest in applying deep learning (DL) approaches that detect patterns from SPECT images, offering automated more accurate methods for detecting PD from brain images, and demonstrating high accuracy in classification between control and PD, as well as between PD stages. Nevertheless, despite their success, these DL-based approaches face challenges in medical adoption due to their "black box" nature, where the decision-making process of the model remains opaque and difficult for clinicians to interpret [8]. To address this gap, recent researches had focused on creating explainable AI models for PD diagnosis. By integrating CNN classifiers with prediction interpreters, such models can identify specific regions of interest in brain images that contribute to the model's predictions, making the decision process more transparent and supporting clinicians in their evaluations [9]. Techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) are being employed to highlight these critical areas, offering more understandable insights into the diagnostic process [10]. This approach promises to improve the reliability and acceptance of DL-based methods in clinical practice by making the outcomes interpretable and actionable.

With the growing aging population and the continued reliance of PD diagnosis on image analysis, applying DL based methods to brain imaging offers the potential to revolutionize early detection and improve patient outcomes, opening new frontiers in the fight against this disease. A study by Suganya Asokan et al. proposes a deep learning approach combining a Hierarchical Spatial Feature CNN with Grad-CAM for the segmentation and classification of Alzheimer's and Parkinson's diseases using MRI scans. The model segments abnormal regions and employs Grad-CAM for explainable classification, achieving a high accuracy of 98.17%, with strong improvements in sensitivity and specificity over existing methods [11]. In [12], Mingjing Yang et al. presented a 3D ResNet model for PD detection using whole-brain MRI scans. The model achieved 96.1% accuracy and 94.5% sensitivity. Grad-CAM highlighted the frontal lobe as key for classification, and UPDRS scores supported the findings, revealing significant differences in language function between PD and control groups.

This paper's goal is to create a DL-based method that uses a 3D CNN along with Grad-CAM to find areas outside of the striatum that help classify PD at different stages using DaTSCAN SPECT imaging, and might be important in early PD diagnosis.

The structure of this paper is as follows: Section 2 details the database collection and preprocessing techniques. In section 3, the experimental methods used as well as the implemented methodology were described. The obtained results from the proposed models were shown in section 4, followed by a discussion of the findings in section 5. Finally, section 6 concludes the study and opens for the new perspectives.

II. DATABASE COLLECTION AND PREPROCESSING

The data used in this study were sourced from the Parkinson's Progression Markers Initiative (PPMI) database [13]. We extracted post-processed DaTSCAN SPECT volumes, each composed of 91 axial slices with a resolution of 109×91 pixels. The dataset comprised 322 healthy controls (HC) volumes, 644 volumes from PD patients at stages 1 and 2, and 81 volumes for stage 3. According to Hoehn and Yahr, Parkinson's disease has five stages; we focused on stages 1 to 3 due to limited data for stages 4 and 5 and to prioritize early detection. To improve model's performance and ensure consistency across samples, we applied two preprocessing techniques to our 3D DaTSCAN volumes: normalization and augmentation. Normalization was performed using min-max scaling to rescale pixel intensity values to a $[0,1]$ range, promoting uniformity and facilitating faster model convergence [14].

After dividing the dataset into three subsets, the train, validation, and test sets, with respective shares of 70%, 20%, and 10%, augmentation was applied on the minority class of the training subset using translation and rotation transformations. These techniques simulate real-world variability in medical imaging such as variations in patient position, scanner angles, and body movements; help the model generalize better by learning from diverse data patterns; improve its performance on unseen data; and by that prevent overfitting [15].

III. EXPERIMENTAL METHODS

The proposed approach applied in this paper to identify PD and categorize subjects as either healthy or diseased is shown in Figure 1. In order to do so, we developed a 3D CNN model for the binary classification of control participants (HC) and PD patients at various stages (1, 2 and 3), and then we generated Grad-CAM heatmaps of the test subsets at each stage.

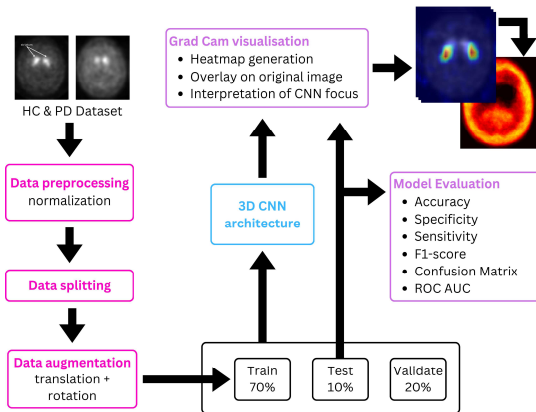


Figure 1: Flowchart of our proposed method

A. 3D Convolutional Neural Network

CNNs are widely used for image analysis due to their ability to automatically extract hierarchical features from raw data. While 2D CNNs are commonly used for traditional image classification, 3D CNNs are ideal for medical imaging, where capturing spatial context and preserving depth information across multiple slices is essential [16].

As seen in Figure 2 below, our 3D CNN is made up of an input layer, three convolutional layers, three pooling layers, three batch normalization layers, five dropout layers, a flatten layer, and two fully connected layers; implemented to gradually extract significant characteristics, reduce dimensionality and avoid overfitting.

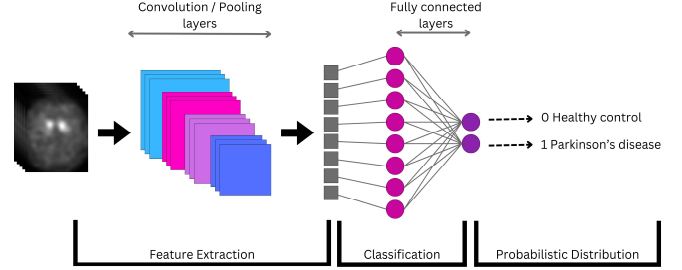


Figure 2: Our proposed 3D CNN architecture

The model processes 3D medical image volumes, where each 3D pixel encodes intensity values obtained from medical imaging techniques. It utilizes three 3D convolutional layers each with increasing filter counts of 32, 64, and 128, and a $3 \times 3 \times 3$ kernel with the ReLU activation function. ReLU is commonly used in CNNs because it introduces non-linearity, preventing the vanishing gradient problem and enabling more efficient and accelerated training [17]. This architecture starts by detecting basic spatial features like edges and textures. As the network deepens, it learns to recognize more abstract patterns and disease-related biomarkers. Throughout the process, spatial locality is preserved, allowing the model to incorporate meaningful context from adjacent voxels [18]. After each convolutional layer, 3D max pooling is used with a $2 \times 2 \times 2$ kernel to reduce spatial resolution, minimize overfitting by removing repetitive information, and emphasize dominant features, preserving critical patterns for accurate classification [18]. Then, batch normalization is applied to stabilize the learning process and accelerate model convergence [19]. Following feature extraction, two fully connected (dense) layers are employed, with 128 and 64 neurons respectively, both activated by ReLU. These layers transform the spatially compressed features into a decision-making process. Dropout layers were added after each convolutional block, with a 0.3 probability rate in the convolutional layers and 0.5 in the dense layers. This technique randomly turns off a subset of neurons during training, helping the model learn more resilient and generalizable features rather than relying on specific noisy patterns [20]. The final output layer, consisting of a single neuron with a sigmoid activation function, returns the probability that a subject has PD or belongs to the control group. The model architecture was selected after extensive trial-and-error experimentation. To promote simplicity and reduce overfitting, L2 regularization was applied to both convolutional and dense layers, alongside early stopping once performance stabilized. Training was carried out over 50 epochs using 10-fold cross-validation to ensure model robustness and generalization.

B. Gradient-weighted Class Activation Mapping :

To analyze the spatial regions that contributed most to the model's classification decisions, we applied Gradient-weighted Class Activation Mapping (Grad-CAM), as introduced in [10], to the final convolutional layer of our 3D CNN. After the prediction is made, the gradients of the output score y^c with respect to the feature maps $A^k \in \mathbb{R}^{D \times H \times W}$ are computed for the target class c . The importance of each activation map is quantified by averaging these gradients across the spatial dimensions:

$$\alpha_k^c = \frac{1}{DHW} \sum_{i=1}^D \sum_{j=1}^H \sum_{l=1}^W \frac{\partial y^c}{\partial A_{i,j,l}^k} \quad (1)$$

These weights α_k^c are then used to compute a class-discriminative heatmap via a linear combination of the feature maps. α_k^c represent how strongly each feature map contributes to the final decision. The final Grad-CAM heatmap $L_{Grad-CAM}^c$ is then obtained as a weighted sum of the feature maps, followed by a ReLU activation to discard negative influences:

$$L_{Grad-CAM}^c = \text{ReLU} \left(\sum_k \alpha_k^c \right) A^k \quad (2)$$

The resulting heatmap $L_{Grad-CAM}^c$ is a 3D volume of the same resolution as the feature maps, which is then interpolated (typically using trilinear interpolation) to match the size of the original input image. This allows for visual superposition and interpretation of which anatomical regions contributed to the classification outcome. The overall process, from activation map extraction to heatmap generation and visualization, is summarized in Figure 3 [21].

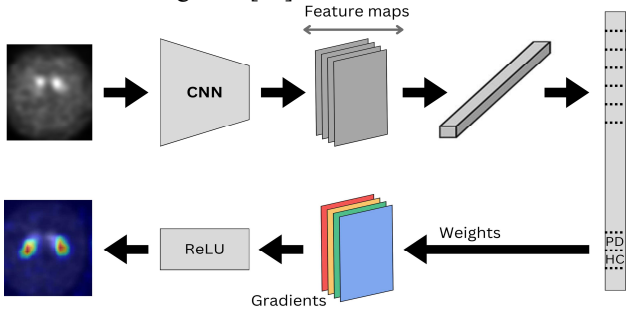


Figure 3: Our Grad-CAM explanatory diagram

IV. RESULTS

Using Google Colab with GPU acceleration, we implemented, trained, and evaluated the proposed model using standard performance metrics. These were derived from key classification outcomes: true positives (TP, correct healthy predictions), true negatives (TN, correct PD predictions), false positives (FP, healthy predicted as PD), and false negatives (FN, PD predicted as healthy). Evaluation metrics included accuracy, sensitivity (recall), specificity (precision), F1-score, confusion matrix, and ROC-AUC, all reported by disease stage in Table 1.

Table 1 Performance evaluation metrics

	Accuracy	Specificity	Sensitivity	F1score
HC-PDs1	98.56%	0.98	0.98	0.98
HC-PDs2	98.34%	0.99	0.99	0.99
HC-PDs3	97.56%	0.98	0.98	0.98

The classification results indicate that the model performs excellently in classifying control and PD subjects at

different stages. Figure 4 represents the plots of both training/validation accuracy and loss for the model classifying control and PD of stage 1. Both training and validation show similar upward trend in the accuracy plot, and their losses decrease steadily indicating that the model is minimizing the errors. These plots indicate that the model is performing well overall.

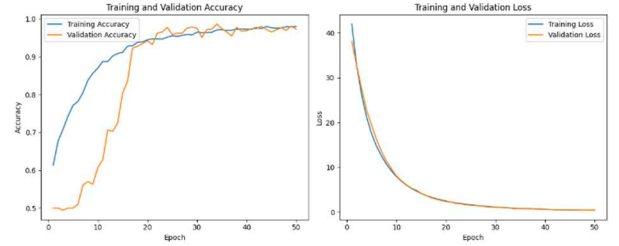


Figure 4: Plots of training/validation accuracy and loss for the model classifying control and PD of stage 1

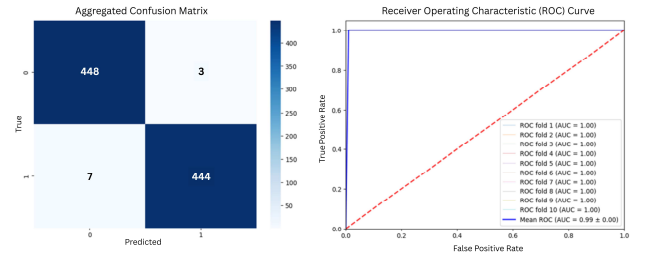


Figure 5: Confusion matrix and ROC-AUC of Control and PD stage 2 model

Figure 5 shows the confusion matrix and ROC-AUC plots of the control/PD stage 2 trained model. The results indicate strong performance as the number of incorrectly predicted classes is very low compared to our number of participants. In addition, the model achieves an area under the curve of 0.99, which indicates perfect performance in distinguishing between the two classes. Moving on to the Grad-CAM results, as seen in figure 6, the heatmaps reveal interesting insights into how the model classifies PD from control subjects. Grad-CAM operates by computing the gradient of the predicted class with respect to the feature maps of a convolutional layer. These gradients indicate how much each spatial location in the feature maps contributes to the final decision.

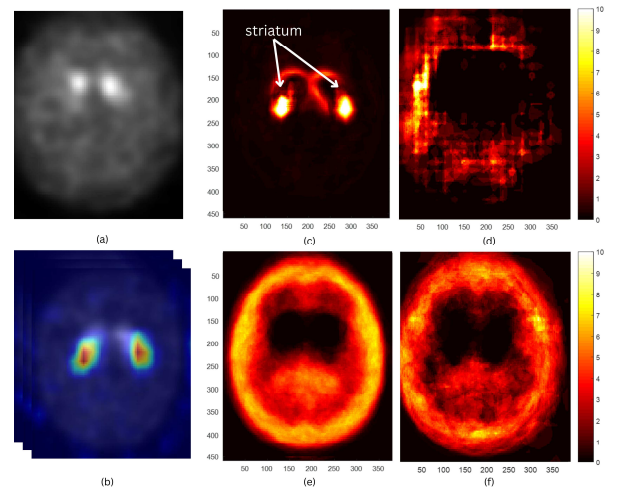


Figure 6: (a) original DaTSCAN image, (b) Grad-Cam heatmap overlaid on the original image, and (c-f) average Grad-CAM maps for control, Parkinson's stages 1, 2 and 3

These findings are further interpreted in the discussion section, where we examine their implications for PD pathophysiology and biomarker discovery.

V. DISCUSSION

Grad-CAM maps are based on computing the gradients of the classification score with respect to the activations of a given convolutional layer. In physical terms, a gradient represents the direction and magnitude of the most rapid variation of a function. In neural networks, this gradient expresses a directional sensitivity of the model output to local variations in the input image. Analyzing these gradients within the context of our 3D CNN classification model provides a deeper understanding of the network's decision-making mechanism. Unlike traditional image enhancement techniques that highlight visually prominent regions, Grad-CAM identifies areas where small changes in signal intensity significantly influence the model's output. These zones of informational instability may correspond to regions of functional vulnerability areas where subtle perturbations disproportionately affect network-level behavior.

This perspective is particularly relevant given the imaging modality used in this study DaTSCAN, which is functional in nature and sensitive to the activity of presynaptic dopaminergic transporters. While it is not a structural technique, the resulting signal can still be modulated by tissue-level physical properties such as synaptic terminal density, neuronal orientation, and local cellular texture. These features, by influencing metabolism or tracer diffusion, shape the image signal and generate subtle patterns detectable by the neural network. The computed gradients then act as indicators of these micro-variations, highlighting regions where fine tissue architecture drives distinct functional responses. The extra-striatal activations observed in our model especially in the cerebellum and prefrontal cortex are particularly noteworthy. While the striatum is traditionally the primary focus in Parkinson's disease diagnosis, our findings suggest that additional brain regions may contribute to early detection. This observation aligns with Braak's [22] staging hypothesis, which describes Parkinson's disease as a progressively ascending pathology that initially affects subcortical structures before involving cortical areas. The regions identified by the model may thus reflect not only functional alterations but also early microstructural anomalies indicative of ongoing neurodegeneration.

VI. CONCLUSION

This study demonstrates the relevance of combining 3D deep learning architectures with explainable artificial intelligence in the domain of functional brain imaging applied to neurodegenerative diseases. The integration of a 3D convolutional neural network with Grad-CAM enabled not only high classification performance but also interpretable neuroanatomical insight into the regions contributing to the model's decisions. Beyond predictive accuracy, the model's ability to highlight extra-striatal regions challenges the conventional diagnostic paradigm focused exclusively on dopaminergic nuclei. These findings support the hypothesis of a broader and more distributed pathophysiological process involving multiple brain circuits even at early stages of the disease. From a clinical application perspective, applying this methodology to other imaging modalities, such as structural MRI, PET, DTI, or fMRI, would help evaluate how the nature

of the imaging signal influences the regions identified as discriminative. A multimodal approach could enable a more integrated mapping of early alterations in Parkinson's disease by combining structural, functional, and metabolic markers. Such integration may contribute to the development of next-generation diagnostic tools that are both high-performing and clinically interpretable.

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